

Assignment 6

1. Suppose that $(X, \mathcal{S})$ is a measurable space. Prove that if $E \in \mathcal{S} \otimes \mathcal{S}$ then

$$\{x \in X : (x, x) \in E\} \in \mathcal{S}$$

Hint: Prove that the collection of elements of $\mathcal{S} \otimes \mathcal{S}$ with this property is a σ -algebra containing the measurable rectangles.

Proof. Let $T = \{E \in \mathcal{S} \otimes \mathcal{S} : \{x \in X : (x, x) \in E\} \in \mathcal{S}\}$. We must show that this set is a σ -algebra containing the measurable rectangles. If we consider the case where E is a measurable rectangle, it must be of the form $A \times B = E$, where $A, B \in \mathcal{S}$. Thus, in order for $(x, x) \in E$, it is necessary that $x \in A$ and $x \in B$. Thus, $x \in A \cap B$. Because A and B are measurable, it must be the case that their intersection, $A \cap B \in \mathcal{S}$. Thus, the diagonal of any measurable rectangle is in $\mathcal{S}$. Thus, T contains the set of measurable rectangles.

□

2. (a) Let a_{mn} be a doubly-indexed sequence of real numbers such that

$$\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} |a_{mn}| < \infty. \text{ Prove that } \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} a_{mn} = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} a_{mn}.$$

Proof. Notice that $\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} |a_{mn}| = \int_{\mathbb{N}} \int_{\mathbb{N}} |a_{mn}| d\mathbf{c}(n) d\mathbf{c}(m)$, when $\mathbf{c}$ is counting measure. Additionally, notice that $(\mathbb{N}, \mathcal{P}(\mathbb{N}), \mathbf{c})$ is σ -finite. Thus we can apply Tonelli's Theorem to $|a_{mn}|$, giving the following:

$$\int_{\mathbb{N}} \int_{\mathbb{N}} |a_{mn}| d\mathbf{c}(n) d\mathbf{c}(m) = \int_{\mathbb{N}} \int_{\mathbb{N}} |a_{mn}| d\mathbf{c}(m) d\mathbf{c}(n) = \int_{\mathbb{N} \times \mathbb{N}} |a_{mn}| d(\mathbf{c} \times \mathbf{c}).$$

We were given that $\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} |a_{mn}| < \infty$, thus $\int_{\mathbb{N} \times \mathbb{N}} |a_{mn}| d(\mathbf{c} \times \mathbf{c}) < \infty$. Therefore we are now able to apply Fubini's Theorem to yield:

$$\begin{aligned} \int_{\mathbb{N} \times \mathbb{N}} a_{mn} d(\mathbf{c} \times \mathbf{c}) &= \int_{\mathbb{N}} \int_{\mathbb{N}} a_{mn} d\mathbf{c}(m) d\mathbf{c}(n) = \int_{\mathbb{N}} \int_{\mathbb{N}} a_{mn} d\mathbf{c}(n) d\mathbf{c}(m). \\ &= \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} a_{mn}. \\ &= \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} a_{mn}. \end{aligned}$$

□

- (b) Find an example of a doubly-indexed sequence a_{mn} of real numbers such that $\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} a_{mn} = 0$ and $\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} a_{mn} = \infty$.

Explain why this is not a contradiction of part (a).

Define a_{mn} by $\{a_m\}_n = 2 - \frac{1}{2^{n-1}}$. Thus, $\sum_{n=1}^{\infty} a_{mn} = 0$, so $\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} a_{mn} = \sum_{m=1}^{\infty} 0 = 0$. However, $\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} a_{mn} = \sum_{n=1}^{\infty} 1 - \sum_{n=1}^{\infty} \frac{1}{2} - \dots = \infty$. This is not a contradiction of (a) because $\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} |a_{mn}| = \infty$, and thus Fubini's theorem does not apply.

3. Let $([0, 1], \mathcal{B}, \lambda)$ be Lebesgue measure with the Borel sets on $[0, 1]$ and $([0, 1], \mathcal{B}, \mu)$ be counting measure with the Borel sets on $[0, 1]$. Consider $\lambda \times \mu$ on $\mathcal{B} \otimes \mathcal{B}$. Prove that $D = \{(x, x) : x \in [0, 1]\} \in \mathcal{B} \otimes \mathcal{B}$ and show that

$$\int_{[0,1]} \int_{[0,1]} \chi_D(x, y) d\lambda(x) d\mu(y) \neq \int_{[0,1]} \int_{[0,1]} \chi_D(x, y) d\mu(y) d\lambda(x)$$

Explain why this is not a contradiction of Fubini or Tonelli's Theorem.

Proof. As we know, $\mathcal{B} \otimes \mathcal{B}$ is the smallest σ -algebra generated by the measurable rectangles on $\mathcal{B} \times \mathcal{B}$. If we can show that $f^{-1}\{0\} = \{(x, x) : x \in [0, 1]\}$ is Borel, then we will have shown that $\{(x, x) : x \in [0, 1]\}$ is measurable and thus that $\{(x, x) : x \in [0, 1]\} \in \mathcal{B} \otimes \mathcal{B}$. Notice that $\{0\}$ is compact, and by continuity the preimage of $\{0\}$ is compact. Compact sets on $\mathbb{R}$ are measurable, and thus $f^{-1}\{0\} \in \mathcal{B} \otimes \mathcal{B}$. We now show that we cannot switch order of integration:

$$\int_{[0,1]} \int_{[0,1]} \chi_D(x, y) d\lambda(x) d\mu(y) = \int_{[0,1]} 0 d\lambda(x) = 0.$$

$$\int_{[0,1]} \int_{[0,1]} \chi_D(x, y) d\mu(y) d\lambda(x) = \int_{[0,1]} 1 d\lambda(x) = 1.$$

This is not a contradiction to Tonelli's Theorem because $([0, 1], \mathcal{B}, \mu)$ is not σ -finite, and thus the theorem does not apply. This is not a contradiction to Fubini's Theorem, because the product measure $\lambda \times \mu$ does not yield a finite integral:

$$\int_{[0,1] \times [0,1]} |\chi_D(x, y)| d(\lambda \times \mu) = \infty.$$

□